

## Research Statement

My long-term goal is to build systems that can *understand, reason, plan, and develop physical intuition* about the world while remaining *efficient, scalable, and controllable*.

Toward **superhuman intelligence** through **efficient, scalable, and controllable** generative and world models.

Toward this goal, my work spans *one-step generative modeling* through *pretraining* (NeurIPS) and *distillation* (ECCV and ICCV), efficient *megapixel-scale synthesis* that roughly halves 4K latency (NeurIPS), and *controllability* and *safety* (ICCV and CVPR), including protection of photos against unauthorized image-to-video generation.

## Education

### Johns Hopkins University

Doctor of Philosophy, Electrical and Computer Engineering

Research: **Generative and World Models**

Aug 2026 – expected 2030

Advisor: Professor Vishal M. Patel

## Research Experience

### Qualcomm AI Research

Predoctoral Researcher

3 yrs 1 mo · Aug 2023 – Aug 2026

- **Six publications**, including **five first-authored**, in generative modeling at ECCV, CVPR, ICLR, 2×NeurIPS, and ICCV.
- **Two U.S. patents**.
- Recipient of the 2025 **Outstanding Resident** in Research and Applied Demo Award, Qualcomm AI Residency Program.
- Advised by **Principal Scientist Dr. Anh Tran**. Completed intensive PhD-level training in a selective research residency, conducting research from problem formulation and experimental design to rigorous evaluation and publication.

## Selected Publications

(\*) denotes equal contribution

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| <b>ECCV</b><br>2026<br>POSTER<br>SLIDES    | <b>Cross-Space Distillation: Teaching One-Step Students with Modern Diffusion Teachers</b><br><b>Anh Nguyen*</b> , Ngan Nguyen*, Duc Vu*, Trung Dao, Viet Nguyen, Quan Dao, Kien Nguyen, Chi Tran, Phong Nguyen, Khoi Nguyen, Cuong Pham, Dimitris Metaxas, Vishal M Patel, and Anh Tran<br>QUALCOMM   JOHNS HOPKINS UNIVERSITY   RUTGERS UNIVERSITY   UNIVERSITY OF WISCONSIN-MADISON |
| <b>NeurIPS</b><br>2026                     | <b>UltraFlash: Accelerating Megapixel Visual Synthesis</b><br>Phuc Lai*, <b>Anh Nguyen*</b> , Phong Nguyen, and Anh Tran<br>QUALCOMM                                                                                                                                                                                                                                                   |
| <b>NeurIPS</b><br>2025<br>POSTER<br>SLIDES | <b>Improved Training Technique for Shortcut Models</b><br><b>Anh Nguyen*</b> , Viet Nguyen*, Duc Vu, Trung Dao, Chi Tran, Toan Tran, and Anh Tran<br>QUALCOMM                                                                                                                                                                                                                          |
| <b>CVPR</b><br>2026<br>POSTER<br>SLIDES    | <b>Anti-I2V: Safeguarding your photos from malicious image-to-video generation</b><br>Duc Vu, <b>Anh Nguyen</b> , Chi Tran, and Anh Tran<br>QUALCOMM                                                                                                                                                                                                                                   |
| <b>ICLR</b><br>2026                        | <b>Revisit Visual Prompt Tuning: The Expressiveness of Prompt Experts</b><br><b>Anh Nguyen*</b> , Minh Le*, Huy Nguyen, Chau Nguyen, Anh Tran, and Nhat Ho<br>QUALCOMM   THE UNIVERSITY OF TEXAS AT AUSTIN                                                                                                                                                                             |
| <b>ICCV</b><br>2025<br>POSTER<br>SLIDES    | <b>Supercharged One-step Text-to-Image Diffusion Models with Negative Prompts</b><br><b>Anh Nguyen*</b> , Viet Nguyen*, Trung Dao, Khoi Nguyen, Cuong Pham, Toan Tran, and Anh Tran<br>QUALCOMM                                                                                                                                                                                        |